

Cybersecurity Incident Report

Section 1: Identify the type of attack that may have caused this network interruption

One potential explanation for the website's connection timeout error message is:DDOS SYN flood attack

The logs show that: The Web server under the ip 192.0.2.1 was receiving and abnormal amount of syn connection requests with out completing the handshake

This event could be: a Malicious ddos tcp syn flood attack

Section 2: Explain how the attack is causing the website to malfunction

When website visitors try to establish a connection with the web server, a three-way handshake occurs using the TCP protocol. Explain the three steps of the handshake:

- 1.SYN a client sends a SYN packet indicating it want to establish a connection with the server

2. SYN/ACK the server sends back the SYN/ACK packer indicating to the client that it can connect

- 3.ACK the client sends an ACK packet signifying the start of a connection

Explain what happens when a malicious actor sends a large number of SYN packets all at once: The server has a table that is stored in the server's ram where it stores the connection requests that starts with "SYN" packets and then after the connection is established the record is deleted form the table.

When a threat actor sends massive amounts of SYN packets the server's ram is filled and resources are depleted

Explain what the logs indicate and how that affects the server: The logs indicate a DDOS. This affects the server by using up all the resources which damages the Availability